

# QDGset : A Large Scale Grasping Dataset Generated with Quality-Diversity

Johann Huber<sup>\*1</sup>, François Hélienon<sup>\*1</sup>, Mathilde Kappel<sup>\*1</sup>, Ignacio de Loyola Paez-Ubieta<sup>\*2</sup>,  
Santiago T. Puente<sup>2</sup>, Pablo Gil<sup>2</sup>, Faïz Ben Amar<sup>1</sup> and Stéphane Doncieux<sup>1</sup>



Universitat d'Alacant  
Universidad de Alicante

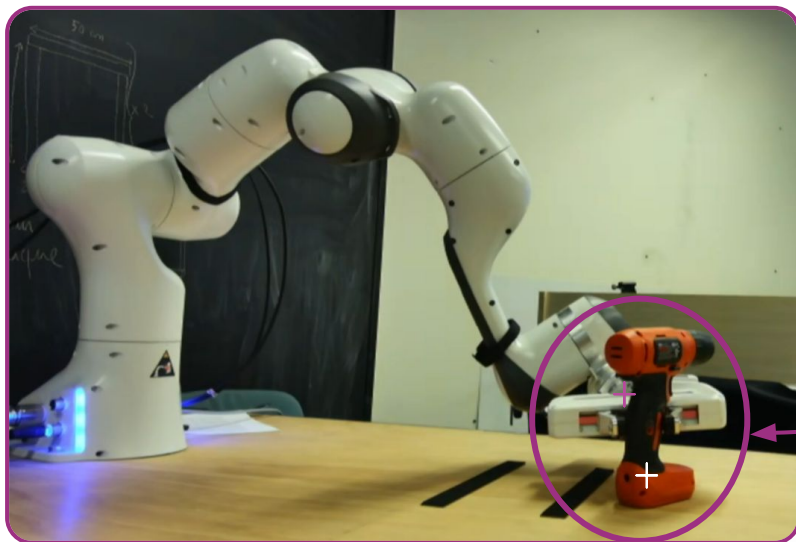


\* equal contribution and corresponding authors

<sup>1</sup>Sorbonne Université, CNRS, Institut des Systèmes Intelligents et de Robotique, ISIR, F-75005 Paris, France

<sup>2</sup>University of Alicante, University Institute for Computing Research (IUII), AUtomatics, Robotics and Artificial Vision research group (AUROVA), E-03690 San Vicente del Raspeig (Alicante), Spain

# Challenge of learning to grasp in robotics



One 6 Degree of Freedom Grasp  
(6DoF grasp)

- $g \in SE(3)$
- Sparse reward task 💎
- No reward  $\rightarrow$  No learning

# Data-driven methods require big datasets

Real world demonstrations  
(limited ❌)



Automatic demonstration  
(unlimited, but challenging ✅)



Deep learning  
approaches

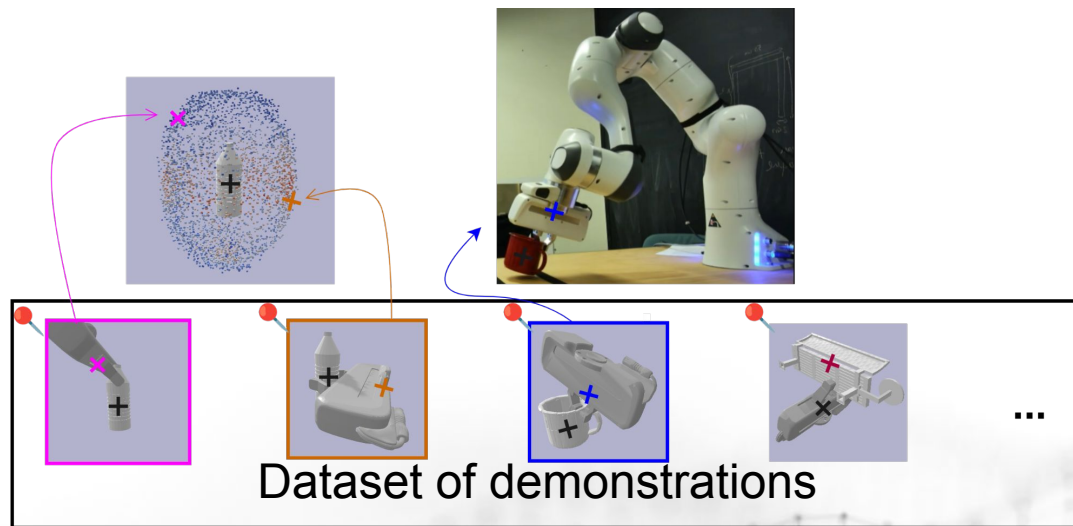


How to automatically  
generate a grasp  
demonstrations  
dataset in a sparse  
reward environment ?



# Problem framing

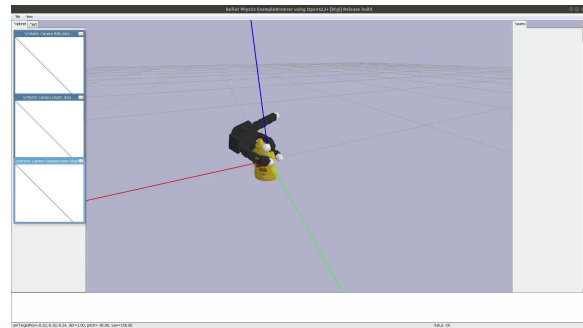
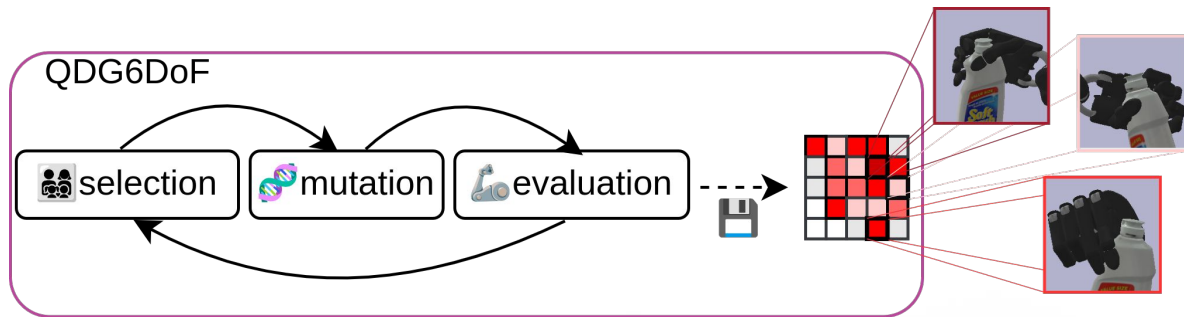
- Dataset of object-centric grasp demonstrations  $g \in SE(3) \times \mathbb{R}^n$
- A demonstration = a 6-DoF end effector pose relative to the object frame
- From 1K to 5K grasps per objects
- Simulated demonstrations transferable in the real world [1]



[1]: Huber, J., H  l  non, F., Watrelot, H., Amar, F. B. & Doncieux, S. (2024). Domain randomization for sim2real transfer of automatically generated grasping datasets

# Generating grasps with Quality-Diversity (QD)

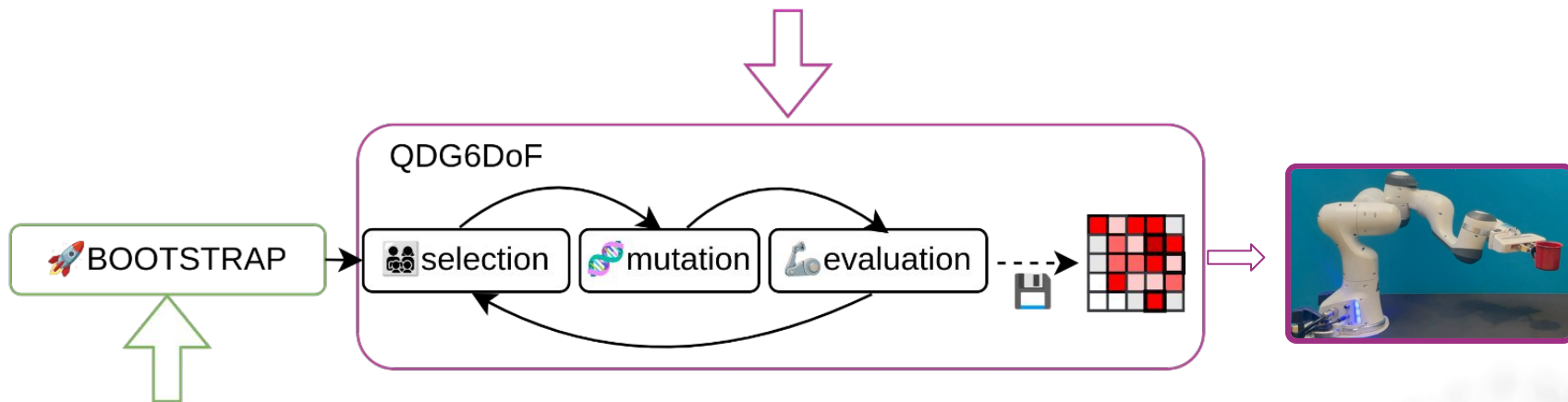
- Efficient exploration of sparse reward environments
- Evolutionary algorithms  $\Rightarrow$  selection-mutation loop
- **Input** = (gripper, object model)
- **Output** = set of diverse and high performing grasp archive



- **QD equations** : 
$$\begin{cases} \forall b \in \mathcal{B}_{reach}, \exists \theta \in A, d_{\mathcal{B}}(\phi_{\mathcal{B}}(\theta), b) < \epsilon & \text{diverse solutions} \\ \forall \theta' \in A, \theta' = \operatorname{argmax}_{\theta \in N(b_{\theta'})} f(\theta) & \text{best-performing solution} \end{cases}$$

# Scale up the production of synthetic grasping datasets

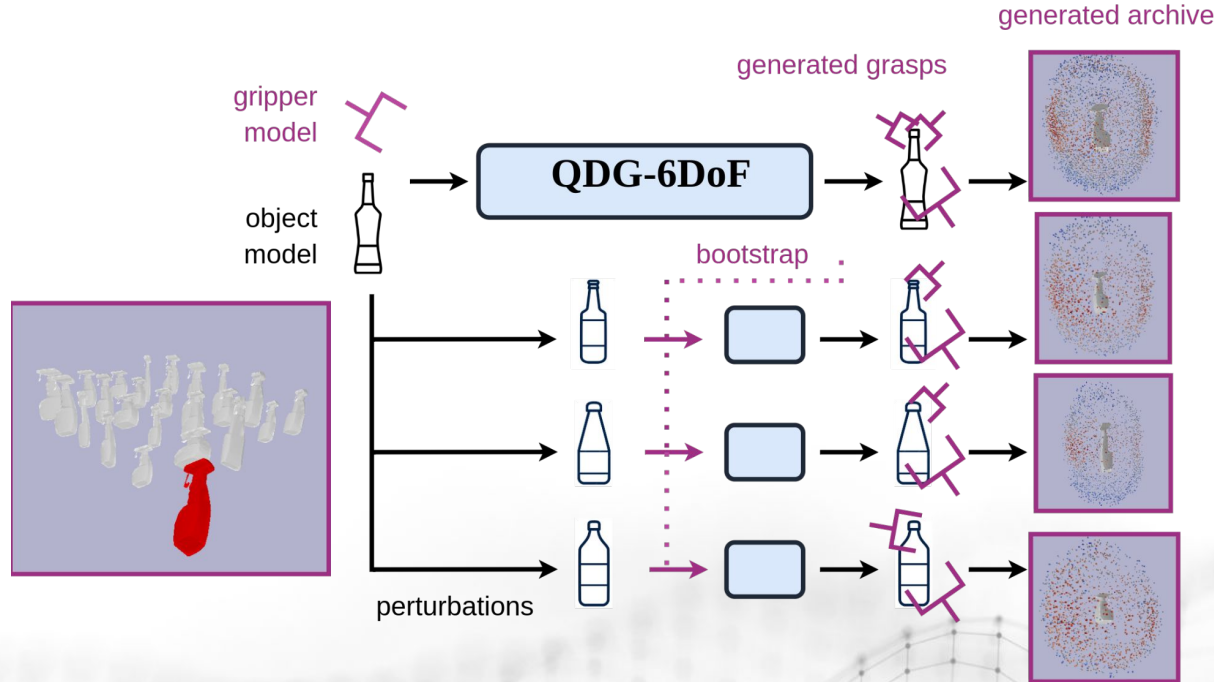
 Huber, J., H  l  non, F., Kappel, M., Chelly, E., Khoramshahi, M., Ben Amar, F., & Doncieux, S. ***Speeding up 6-DoF Grasp Sampling with Quality-Diversity.*** (IROS'24)



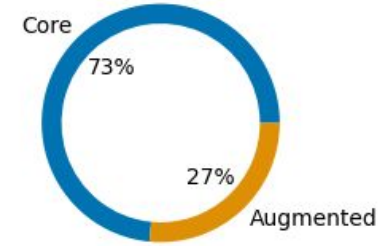
  Huber, J., H  l  non, F., Kappel, M., P    -Ubieta, I. de L., Puente, S. T., Gil, P., Ben Amar, F., & Doncieux, S. ***QDGset: A Large Scale Grasping Dataset Generated with Quality-Diversity.*** Sorbonne Universit   / University of Alicante.

# Bootstrap speeds up and scales up demonstrations 🚀

- Run QD optimisation module on a given object  $\Rightarrow$  generate  $A_s^b$
- Use  $A_s^b$  to bootstrap the optimisation on similar objects



# Object data augmentation



- Core initial object datasets  $P$ ,  $n = \mathbf{29\ 651\ objects}$

$$P = (\mathcal{P}_i)_{1 \leq i \leq n} \quad \text{where } \mathcal{P}_i = \{p_j \in \mathbb{R}^3 \mid p_j = (x_j, y_j, z_j), j = 1, 2, \dots, m_i\}$$

- Data augmentation = point cloud deformation  $\varphi D_a$

$$\mathcal{P}_i^a = \{\varphi_{D_a}(p_j) = D_a \cdot p_j \mid p_j \in \mathcal{P}_i\} \quad \text{where } D_a = \text{DIAG}(\alpha_1, \alpha_2, \alpha_3)$$

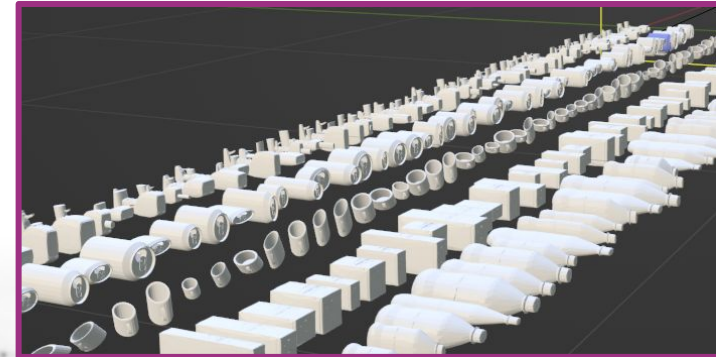
- Final dataset  $\Rightarrow$  size  $\mathbf{40\ 353\ objects}$**

$$P' = (\mathcal{P}_i^a)_{1 \leq i \leq n, 1 \leq a \leq a_{\max}}$$

- QDGset is  $\mathbf{4\ times}$  bigger than ACRONYM [2]

62M grasps

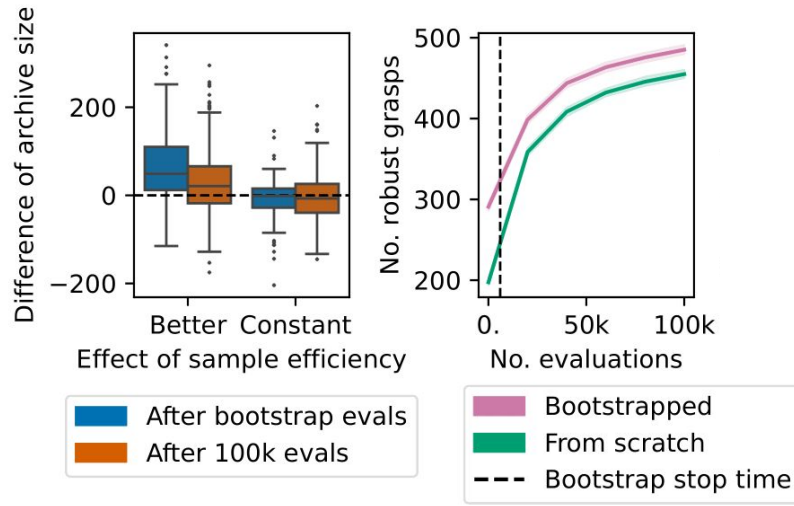
17.7M grasps



[2] : Eppner, C., Mousavian, A. & Fox, D. (2021). Acronym: A largescale grasp dataset based on simulation. In 2021 IEEE International Conference on Robotics and Automation (ICRA) (pp. 6222–6227)

# Results of bootstrap on augmented data

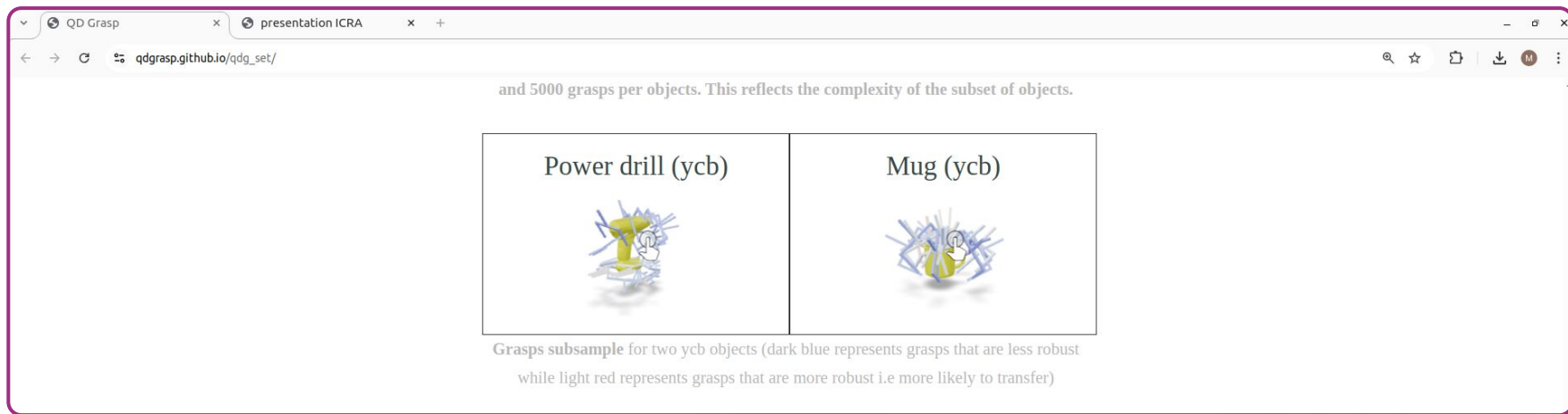
- Bootstrap reduces the required number of evaluations by up to **20%**
- Better performances when stopping the optimization right after the bootstrap



- Bootstrapping does maintain the quality of the grasps

# Conclusions on QDGset

- Framework for scaling up object-centric poses
- 62M grasps on about 40K objects
- Allow anyone to easily produce data they need



[https://qdgrasp.github.io/qdg\\_set/](https://qdgrasp.github.io/qdg_set/)



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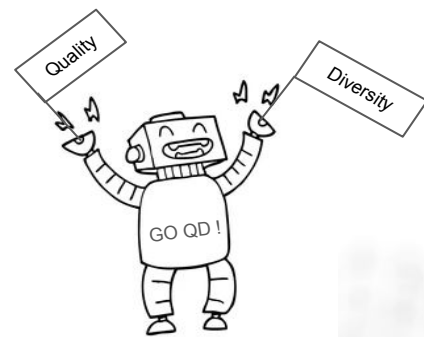
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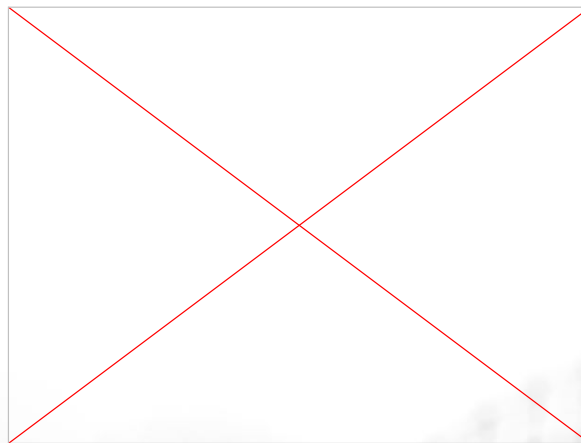
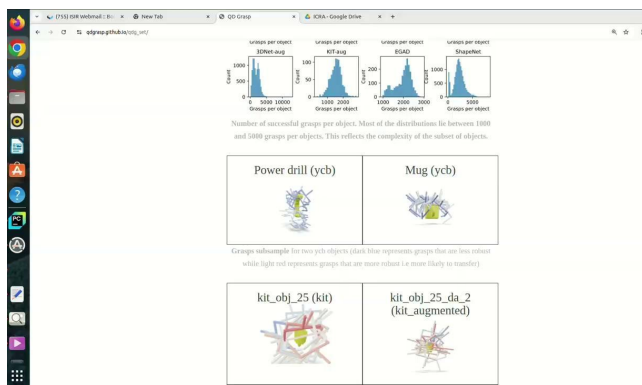
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# Conclusion

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